Roll No. | | | | | | | |

End Semester Examination 2025

Name of the Course: B.Tech (ECE) Semester: VI Sem

Name of the Paper: Microwave Engineering Paper Code: **TEC 602**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Explain the various modes of propagation in Electromagnetic Wave propagation.	
(b)	Derive the field equation for rectangular waveguide. Why TEM mode cannot exist in rectangular waveguide?	**CO 1**
(c)	A rectangular waveguide with dimensions a = 2.29 cm and b = 1.02 cm is used to transmit microwaves. The operating frequency is 10 GHz. Determine the dominant mode of propagation and whether this frequency will allow propagation of higher-order modes.	
Q2	(20 marks)	
(a)	What is a scattering matrix in microwave communication systems? Explain its significance.	
(b)	Explain the working of isolator and circulator with the scattering matrix.	**CO 2**
(c)	A 3 dB, 2-way power divider is used to split an input power of 20 W. Calculate the power delivered to each output port and the voltage at each output port.	
Q3	(20 marks)	
(a)	Explain in detail about 2-cavity klystron amplifier with diagram. Also explain the bunching process.	
(b)	Explain the principle of operation of Tunnel diode. Give the V-I characteristics with the help of energy band diagram.	**CO 3**
(c)	A Gunn diode operates at a frequency of 10 GHz, with a bias voltage of 10 V. The diode has a current of 50 mA and the efficiency of the Gunn diode is 40%. Calculate the power output of the Gunn diode	
Q4	(20 marks)	
(a)	What are the various methods for measuring VSWR? Discuss them in detail.	
(b)	Explain the spectrum analyzer with detailed block diagram. What are the various applications of spectrum analyzer?	
(c)	A microwave power measurement system consists of a directional coupler connected to a power sensor. The coupling factor of the directional coupler is 20 dB, and the power measured at the coupled port is 1 mW. Calculate: (a) The incident power on the main line. (b) The incident power in dBm. (c) If the reflected power measured at another coupled port is 0.01 mW, calculate the return loss (in dB).	**CO 4**

Q5	(20 marks)	**CO 5, 6**
(a)	Explain in detail the Richard's transformation and Kurodas identities.	
(b)	What are the Key Concepts in Low-Pass Prototype Design.	
(c)	Design a maximally flat low pass filter using lumped elements at f_c=2 GHz and of order 5. Transform the filter low pass filter to high pass filter and band pass filter with f_c=2 GHz and 10% fractional bandwidth. Z_0=50 Ω $g_1 = g_5 = 0.618$; $g_2 = g_4 = 1.618$; $g_3 = 2.0$	